Project Chimera Intelligent Core

1. Project Name

Project Chimera Document Intelligence System for Windows 11

2. Objective

The primary objective of Project Chimera is to provide offline-capable, privacy-first semantic document question answering on Windows 11 workstations using local vector embeddings and low-latency LLM inference. It enables enterprise users to query proprietary technical manuals, legal reports, and research papers without leaking corporate data.

3. Architecture

The system employs a 5-stage pipeline: (1) Ingestion with format-specific extraction for PDF, DOCX, TXT, CSV, and Markdown; (2) Semantic recursive chunking using an 800-character window with 100-character overlap; (3) 384-dimensional dense embeddings generated locally by SentenceTransformers all-MiniLM-L6-v2; (4) Persistent ChromaDB vector indexing with cosine distance metric; (5) Strict prompt-constrained LLM generation powered by Google Gemini.

4. Technology Stack

Backend: Python 3.11+, FastAPI, Uvicorn, python-dotenv.
Vector Database: ChromaDB persistent client.
Embeddings: Sentence-Transformers (all-MiniLM-L6-v2).
Frontend: React 18, Vite, Lucide Icons, Windows 11 Fluent UI CSS.
LLM Provider: Google Gemini API (gemini-2.5-flash).

5. Advantages

Enterprise privacy with 100% local embedding calculation and on-premise vector storage. Near-instantaneous semantic retrieval under 50 milliseconds. Distinctive Windows 11 Fluent aesthetic featuring a live Retrieval Inspector for full RAG auditing.

6. Limitations

Scanned image-only PDFs without an OCR layer require external pre-processing. Designed primarily for single-workstation deployment rather than distributed cloud clusters.

7. Future Scope

Hybrid lexical and dense search via BM25 re-ranking, multi-modal tabular analysis, and support for local quantized Ollama inference in air-gapped secure Windows environments.